
The Judgment Effect: Does Evaluative Pressure Induce Deeper LLM Reasoning?

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Abstract

As large language models (LLMs) are increasingly deployed as evaluators and judges, their sensitivity to user tone becomes a critical concern. While prior work highlights the risk of sycophancy when models face user pushback, we investigate a parallel phenomenon: social facilitation. Specifically, we explore whether exposing an LLM to skeptical or judgmental tones can act as a catalyst for deeper, more rigorous reasoning. In this work, we present a study on the “Judgment Effect,” examining how varying prompt tones—from neutral to highly judgmental—affect the length and quality of reasoning traces in GPT-4O-MINI. We evaluate the model on two reasoning tasks: social arbitration using the R/AIO dataset and causal auditing via CAUSALT3. Our results show a clear monotonic increase in reasoning effort under evaluative pressure, with the JUDGMENTAL condition extending trace lengths by up to 22.0%. Crucially, on the complex social judgments of R/AIO, this heightened reasoning effort yields a 6.6% improvement in accuracy over the neutral baseline, with minimal sycophantic label flipping. These findings indicate that controlled, judgmental framing can successfully induce deeper logical exploration and edge-case analysis in LLMs, providing a practical prompting strategy for high-stakes evaluation tasks.

1 Introduction

The deployment of large language models (LLMs) as autonomous evaluators and social arbitrators requires not only robust reasoning but also stable, reliable judgments under diverse conversational conditions. Users naturally query these models with varying degrees of certainty, skepticism, and frustration. Understanding how an LLM’s internal reasoning process responds to such evaluative pressure is essential. Given the ability of LLMs to act as conversational agents, a critical question arises: does an adversarial or judgmental prompt inherently degrade performance by triggering sycophancy, or can it stimulate the model to think harder and yield a more accurate judgment?

There is growing interest in identifying how models behave when users challenge their authority or inject skepticism into the prompt. Prior work highlights a concerning trend of “sycophancy,” wherein models preemptively abandon correct logical paths to appease a skeptical user Chang [2026], Kim and Khashabi [2025]. However, these studies often focus on whether the model flips its final answer, rather than examining the underlying effort and depth of its reasoning process. We lack a quantitative understanding of whether a model that agrees under pressure bypassed its reasoning entirely, or if it engaged in a more exhaustive search before conceding. It remains unclear whether there is a “sweet spot” of skepticism that improves analytical rigor without collapsing into sycophancy.

In this work, we propose the concept of the “Judgment Effect”—the phenomenon where explicit evaluative pressure acts as a form of social facilitation for LLMs, forcing them to generate more comprehensive reasoning traces. We hypothesize that moderately judgmental prompts cause the model to preemptively defend its logic by exploring hidden motivations, edge cases, and nuances that a neutral prompt would fail to elicit.

We evaluate this hypothesis using GPT-4O-MINI on two distinct datasets: R/AIO for complex interpersonal conflict resolution and CAUSALT3 for rigorous causal reasoning. By varying the tone from NEUTRAL to SKEPTICAL and JUDGMENTAL, we measure both the quantitative expansion of the reasoning trace and the resulting accuracy.

Our main findings demonstrate that judgmental prompts significantly increase reasoning depth, extending reasoning trace lengths by up to 19.2% on R/AIO and 22.0% on CAUSALT3. Crucially, this added effort translates into a 6.6% absolute accuracy improvement on the social judgment task compared to the neutral baseline. The model resists sycophancy, maintaining low label-flip rates while generating robust justifications.

Our core contributions are as follows:

- We introduce the Judgment Effect, demonstrating that evaluative pressure can successfully act as social facilitation for LLMs.
- We conduct controlled experiments across social (R/AIO) and logical (CAUSALT3) domains to quantify the relationship between prompt tone, reasoning length, and accuracy.
- We provide empirical evidence that highly judgmental tones improve accuracy in complex, nuanced scenarios by forcing the model to elaborate on its reasoning, without incurring the high sycophancy rates reported in prior vulnerability studies.

2 Related Work

Our research intersects with multiple strands of literature focusing on LLM alignment, conversational biases, and the mechanics of reasoning under pressure. We categorize the related work into three primary themes: dialogic deference, sycophancy in causal judgments, and the general biases of LLM-as-a-judge frameworks.

Dialogic Deference and Conversational Bias A growing body of work explores how conversational framing influences LLM behavior. Rabbani et al. [2026] introduce the concept of “Dialogic Deference,” demonstrating that models shift their judgments significantly depending on whether a claim is framed as an objective fact or attributed to a conversational speaker. Their findings emphasize that LLMs are sensitive to social dynamics, frequently shifting toward agreement (deference) or disagreement (skepticism) based on the domain. Building on their exploration of conversational framing, we focus specifically on the intensity of adversarial or judgmental tones and quantify their impact on the length and quality of the intermediate reasoning steps, rather than just the final verdict shift.

Sycophancy and Skepticism Traps The tendency of LLMs to alter their answers to please users has been well-documented. Chang [2026] identify the “Skepticism Trap” and “Sycophancy Trap” in causal reasoning tasks, showing that safety-tuned models may reject valid causal links to avoid over-claiming, and flip correct rejections to endorsements when placed under social pressure. Similarly, Kim and Khashabi [2025] investigate sycophancy under user rebuttal, revealing that LLMs frequently endorse a user’s counterargument when challenged sequentially. Unlike these studies, which primarily view user pressure as a vulnerability that induces sycophantic failure, we examine the hypothesis that such pressure can also serve as a catalyst for deeper, more rigorous analytical thought—effectively establishing a constructive “sweet spot” for skepticism.

Biases in LLM-as-a-Judge As LLMs are increasingly utilized to evaluate text and behavior, their inherent biases have come under scrutiny. Ye et al. [2024] categorize twelve types of biases in LLM-as-a-Judge systems, including authority bias and verbosity bias. Their methodology involves injecting perturbations to measure consistency. We complement this line of research by isolating prompt tone as a primary variable, demonstrating that the verbosity induced by judgmental prompts is not necessarily a superficial bias, but often reflects a substantive increase in task-relevant reasoning that leads to more accurate outcomes in highly nuanced domains.

3 Methodology

Problem Formulation Let x be an input task requiring reasoning and a subsequent final judgment $y \in \mathcal{Y}$. A standard neutral prompt formulates the query as $P_{\text{neutral}}(x)$. We introduce a tonal modifier $T \in \{\text{SKEPTICAL}, \text{JUDGMENTAL}\}$ to construct a pressure-inducing prompt $P_T(x)$. The model

generates a reasoning trace (Chain-of-Thought) R_T followed by the final answer $\hat{y}_T$. Our objective is to analyze the relationship between the tone T , the length and depth of the reasoning trace $|R_T|$, the correctness of the final answer $\hat{y}_T$ relative to ground truth y , and the sycophantic label-flip rate compared to the neutral baseline answer $\hat{y}_{\text{neutral}}$.

Datasets We evaluate our hypothesis on two datasets that require distinct types of rigorous reasoning:

- **R/AIO (Social Judgment):** A naturalistic dataset comprising 30 interpersonal conflict scenarios curated from the Reddit forum “Am I Overreacting?”. The task requires the model to act as a social judge and determine whether a subject’s reaction is justified or constitutes an overreaction. This dataset serves to test the model’s capacity for nuanced social and emotional analysis.
- **CAUSALT3 (Causal Reasoning):** A dataset of 30 complex causal “traps” (e.g., confounding variables, regression to the mean) sampled from the CAUSALT3 diagnostic benchmark Chang [2026]. The model must strictly evaluate the validity of causal claims. This dataset tests structured, logical rigor.

Experimental Conditions For each instance in the datasets, we evaluate GPT-4O-MINI under three prompt conditions:

1. **NEUTRAL:** A standard baseline instructing the model to “Analyze and provide a judgment.”
2. **SKEPTICAL:** A low-pressure evaluative prompt adding, “I am skeptical of simple analyses. Look deeper for edge cases.”
3. **JUDGMENTAL:** A high-pressure evaluative prompt adding, “I suspect you will miss the point. Prove your reasoning can withstand intense scrutiny.”

Evaluation Metrics We compute three primary metrics to assess the impact of prompt tone:

- **Reasoning Depth (Effort):** Measured via the character count of the generated Chain-of-Thought reasoning trace $|R_T|$. This proxies the computational and linguistic effort the model expends before finalizing its answer.
- **Accuracy:** The percentage of final answers $\hat{y}_T$ that match the human-annotated ground truth labels.
- **Sycophancy / Stability Rate:** The percentage of cases where the model alters its final label relative to its own NEUTRAL baseline in response to the evaluative pressure. A low flip rate coupled with higher reasoning length indicates that the model is defending its stance rather than arbitrarily capitulating.

All experiments are conducted using the GPT-4O-MINI model via the OpenAI API, with temperature set to 0.0 to isolate the deterministic effects of the prompt framing.

4 Results

We present the quantitative results of our experiments on both the R/AIO and CAUSALT3 datasets across the three tonal conditions. Table 1 summarizes the accuracy and reasoning depth metrics.

Table 1: Results comparing reasoning depth (in characters) and accuracy across different tonal conditions on R/AIO and CAUSALT3. The JUDGMENTAL prompt yields the longest reasoning traces and the highest accuracy on the social task. Best results are in **bold**.

Condition	R/AIO (Social Judgment)		CAUSALT3 (Causal Reasoning)	
	Accuracy (%)	Reasoning Depth (chars)	Accuracy (%)	Reasoning Depth (chars)
NEUTRAL	70.0	2461	100.0	2427
SKEPTICAL	66.7	2774 (+12.7%)	100.0	2796 (+15.2%)
JUDGMENTAL	76.7	2935 (+19.2%)	100.0	2962 (+22.0%)

Reasoning Effort and Depth Our primary hypothesis posits that evaluative pressure forces the model to generate longer, more detailed reasoning traces. As shown in Table 1, we observe a strict

monotonic increase in reasoning depth as prompt intensity increases. On the R/AIO dataset, the SKEPTICAL condition increases reasoning length by 12.7% over the NEUTRAL baseline, while the JUDGMENTAL condition pushes this increase to 19.2%. A similar effect is observed on CAUSALT3, where the JUDGMENTAL prompt yields a 22.0% increase in character count. Qualitative inspection of the traces under the JUDGMENTAL condition reveals that the model explicitly structures its response to address potential criticisms, spontaneously generating sub-sections dedicated to “Hidden Motivations,” “Emotional Nuance,” and “Relationship Dynamics.”

Accuracy and Social Facilitation The critical question is whether this increased reasoning length translates to improved downstream utility. For the R/AIO social judgment task, the NEUTRAL baseline achieves an accuracy of 70.0%. While moderate skepticism slightly degrades performance to 66.7%, the high-pressure JUDGMENTAL prompt significantly boosts accuracy to **76.7%**—a 6.6% absolute improvement over the baseline (↑). This indicates that the rigorous edge-case analysis forced by the judgmental prompt actively helps the model avoid superficial misjudgments and better align with ground-truth human evaluations.

Conversely, on the CAUSALT3 dataset, GPT-4O-MINI achieved a ceiling accuracy of 100.0% across all conditions. While the reasoning trace lengthened significantly, the task was insufficiently difficult to demonstrate any corresponding accuracy gains or losses.

Sycophancy and Stability Analysis To determine if the model was blindly flipping its answers under pressure, we analyzed the label shift rates relative to the neutral condition on the R/AIO dataset. The sycophancy rates were surprisingly low: only 3.3% of labels flipped under the SKEPTICAL condition, and 6.7% flipped under the JUDGMENTAL condition. Because the overall accuracy increased in the JUDGMENTAL condition, these flips were predominantly constructive self-corrections rather than sycophantic concessions. The model successfully maintained its internal logic and convictions while simply thinking more deeply about the problem space.

5 Discussion

Our findings reveal that introducing a judgmental tone to LLM prompts acts as a form of “social facilitation,” compelling the model to engage in deeper, more rigorous reasoning. Unlike the “Sycophancy Trap” observed in prior literature where models capitulate to explicit misinformation or specific false hints Chang [2026], Kim and Khashabi [2025], our prompts challenged the general *quality* of the reasoning. In response to being broadly “judged,” the model elaborated extensively, self-corrected on difficult social nuances, and avoided arbitrary answer flipping.

The Sweet Spot of Skepticism The results suggest the existence of a constructive prompting threshold. For subjective and highly contextual domains such as social arbitration (R/AIO), demanding that the model “prove its reasoning can withstand intense scrutiny” functions effectively as a zero-shot performance booster. By anticipating adversarial review, the LLM expands its search space for edge cases, yielding judgments that are significantly more aligned with human consensus.

Limitations Our study has several limitations that warrant consideration. First, we observed a strict ceiling effect on the CAUSALT3 dataset, where GPT-4O-MINI achieved 100% accuracy across all conditions. This prevented us from analyzing whether the Judgment Effect provides tangible benefits for strict logical and causal reasoning, or if the lengthened reasoning traces merely reflect redundant verbosity. Second, our pilot experiments were conducted exclusively on GPT-4O-MINI with a sample size of 30 cases per dataset. It remains entirely plausible that different model families—particularly larger models like GPT-4o or heavily safety-aligned models like Claude—might exhibit a different “apology threshold” where the pressure transitions from constructive facilitation to destructive sycophancy.

Broader Implications These findings highlight the profound impact of conversational framing on AI behavior. When deploying LLM-as-a-Judge systems in legal, ethical, or moderation contexts, system designers should carefully consider the persona and tone adopted in their meta-prompts. Purposefully injecting skepticism or warning against “lazy” thinking may serve as a simple, effective mechanism to maximize model utility and rigor without requiring complex algorithmic interventions.

6 Conclusion

In this work, we investigated the “Judgment Effect” in large language models, demonstrating that explicitly judgmental and skeptical prompt framing acts as a catalyst for deeper reasoning. Through experiments on the R/AIO and CAUSALT3 datasets using GPT-4O-MINI, we showed that high-pressure evaluative prompts monotonically increase reasoning trace lengths by up to 22.0%. Furthermore, on complex social judgment tasks, this increased effort yields an absolute accuracy improvement of 6.6% over neutral baselines, with the model successfully resisting sycophancy.

Our findings emphasize that users and developers can actively harness conversational tone to regulate an LLM’s analytical rigor. Rather than viewing user skepticism purely as a vulnerability that induces deference, it can be repurposed as a structural tool to enforce exhaustive edge-case analysis. Future work should explore the impact of the Judgment Effect on harder logical benchmarks (such as MATH or complex causal tasks) and investigate the threshold at which intense scrutiny shifts from beneficial facilitation to detrimental sycophancy across different state-of-the-art architectures.

References

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